

Conduct

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Grad IO

Conduct Testing in Industrial Organization

Foundational Empirical IO Question: How do we observe data on price and quantity and infer which model of firm behavior generated those outcomes?

- ▶ Early work: Porter (1983), Bresnahan (1982,1987)
- ▶ Subsequent work defined the “menu” approach: Nevo (1998, 2001), Villas-Boas (2007)
- ▶ Recent revival of “internalization” parameters: Miller and Weinberg (2017), Crawford, Lee, Whinston, and Yurukoglu (2017), Pakes (2017)
- ▶ Parallel work by: Duarte, Magnolfi, Sølvesten, Sullivan (2022) which test is best (RV). Magnolfi, Quint, Sullivan, Waldfogel (2022) Should we test or estimate?
- ▶ Applications of our test: Starc and Wollman (2022), Scuderi (2022), others?

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Is conduct testable? Berry and Haile (2014): yes.

- ▶ Absent additional restrictions, we cannot generally look at data on (P, Q) and decide whether or not collusion is taking place.
 - You say we started colluding at date t , I say we received a correlated shock to mc .
- ▶ We can make progress in two ways: (1) parametric restrictions on marginal costs; (2) exclusion restrictions on supply.
 - Most of the literature focuses on (1) by assuming something like: $\ln mc_{jt} = x_{jt}\gamma_1 + w_{jt}\gamma_2 + \omega_{jt}$.
 - In principle (2) is possible if we have instruments that shift demand for products but not supply. (These are much easier to come up with than “supply shifters”).

A famous plot (Bresnahan 87)

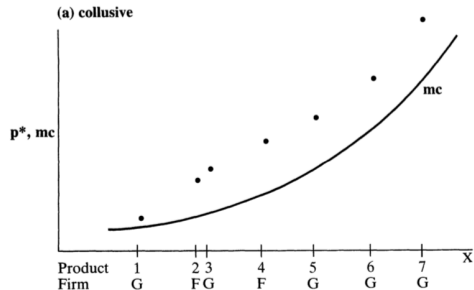


Figure 2(a)

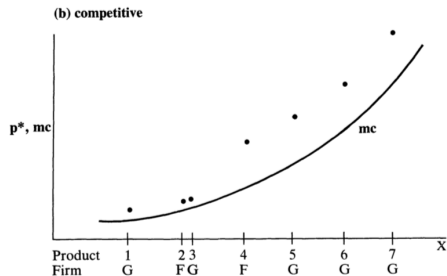


Figure 2(b)

Bresnahan (1980/1982) recognized this problem: we need “rotations of demand”.

Conduct Testing in Pictures (Berry Haile 2014)

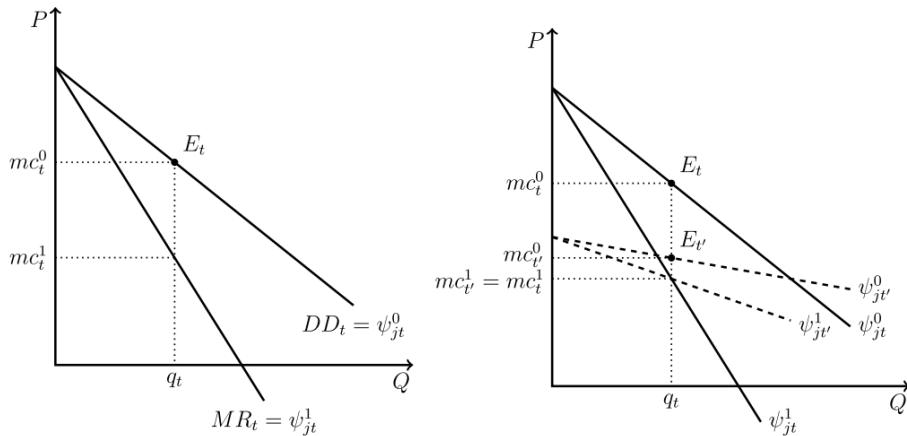


Figure 2(ab) from Berry and Haile (2014), Example 1.

We begin with a relatively standard BLP-style differentiated products setup.

- ▶ Markets t
- ▶ Products j
- ▶ Data $\chi_t = \{(x_{jt}, v_{jt}, w_{jt}) \text{ for all } j \in \mathcal{J}_t\}$.
- ▶ Market Shares $\mathbf{s}_t = [s_{1t}, \dots, s_{Jt}, s_{0t}]$.
- ▶ Prices $\mathbf{p}_t = [p_{1t}, \dots, p_{Jt}]$.
- ▶ Consumers i with demographics y_{it} (income, presence of kids)

Testing Conduct: Setup

We generalize the $\mathcal{H}(\kappa)$ and derive multi-product Bertrand FOCs:

$$\begin{aligned}\arg \max_{\mathbf{p} \in \mathcal{J}_f} \pi_f(\mathbf{p}) &= \sum_{j \in \mathcal{J}_f} (p_j - c_j) \cdot q_j(\mathbf{p}) + \kappa_{fg} \sum_{j \in \mathcal{J}_g} (p_j - c_j) \cdot q_j(\mathbf{p}) \\ \rightarrow 0 &= q_j(\mathbf{p}) + \sum_{k \in (\mathcal{J}_f, \mathcal{J}_g)} \kappa_{fg} \cdot (p_k - c_k) \frac{\partial q_k}{\partial p_j}(\mathbf{p})\end{aligned}$$

- Instead of 0's and 1's we now have $\kappa_{fg} \in [0, 1]$ representing how much firm f cares about the profits of g .
- If f and g merge (or fully coordinated) then $\kappa_{fg} = 1$
 - Often in the real world firms cannot reach fully collusive profits and $\kappa_{fg} \in (0, 1)$.
 - Evidence that $\kappa_{fg} > 0$ is not necessarily evidence of malfeasance, just a deviation from **static Bertrand pricing**

Testing Conduct: Setup

- ▶ Recall the Δ matrix which we can write as $\Delta = \tilde{\Delta} \odot \mathcal{H}(\kappa)$, where $\odot$ is the element-wise or Hadamard product of two matrices.
 - $\tilde{\Delta}$ is the matrix of demand derivatives with $\Delta(j, k) = \frac{\partial q_j}{\partial p_k}$ for all elements.
 - $\mathcal{H}(\kappa) = \kappa_{fg}$ for products owned by (f, g) where $\kappa_{ff} = 1$ always.
- ▶ Mergers are about changing 0's to 1's in the $\mathcal{H}(\kappa)$ matrix.
- ▶ Matrix form of FOC: $q(\mathbf{p}) = \Delta(\mathbf{p}, \kappa) \cdot (\mathbf{p} - \mathbf{mc})$
- ▶ $\mathbf{mc} = \mathbf{p} - \underbrace{\Delta(\mathbf{p}, \theta_2, \kappa)^{-1} \boldsymbol{\sigma}(\mathbf{p})}_{\eta(\mathbf{p}, \mathbf{s}, \theta_2, \kappa)}$ where η_{jt} is the markup.

Reasons for Deviations from Static Bertrand

Biased estimates of own and cross price derivatives: For anything to work, you have correct estimates of $\tilde{\Delta}$. My prior is most papers **underestimate** diversion ratios for close substitutes.

Vertical Relationships: Who sets supermarket prices? Just the retailer? Just the manufacturer? Some combination of both? Retailers tend to **soften** downstream price competition.

Faulty Timing Assumptions: Bertrand is a simultaneous move pricing game. Lots of alternatives (Stackelberg leader-follower, Edgeworth cycles, etc.).

Dynamics and Dynamic Pricing: Forward looking firms or consumers might not set static Nash prices. [e.g. Temporary Sales, Switching Costs, Network Effects, etc.]

Unmodeled Supergame: Maybe firms are legally tacitly colluding, higher prices might be about what firms believe will happen in a price war.

Assume additivity, and write in terms of structural errors:

$$\begin{aligned}\delta_{jt}(\mathbf{s}_t, \tilde{\theta}_2) + \alpha p_{jt} &= h_d(\mathbf{x}_{jt}, \mathbf{v}_{jt}, \theta_1) + \xi_{jt} \\ f(p_{jt} - \eta_{jt}(\mathbf{p}, \mathbf{s}, \theta_2, \kappa)) &= h_s(\mathbf{x}_{jt}, \mathbf{w}_{jt}, \theta_3) + \omega_{jt}\end{aligned}$$

- ▶ To simplify slides we let $f(x) = x$ (often $f(x) = \log(x)$) but we can put that in $h_s(\cdot)$.
- ▶ $h(\cdot)$ are often just linear relationships like: $\theta_1[\mathbf{x}_{jt}, \mathbf{v}_{jt}]$.
- ▶ Endogeneity Problem: p_{jt} and η_{jt} are functions of (ξ, ω) .
- ▶ (θ_2, κ) parameters that determine markups

Approach #1: Demand Side

1. Estimate θ_2 from demand alone.

$$\delta_{jt}(\mathbf{s}_t, \tilde{\theta}_2) + \alpha p_{jt} = h_d(\mathbf{x}_{jt}, \mathbf{v}_{jt}, \theta_1) + \xi_{jt}$$
$$E[\xi_{jt} | \mathbf{x}_t, \mathbf{v}_t, \mathbf{w}_t] = 0$$

2. Recover marginal costs $\widehat{\mathbf{mc}} = \mathbf{p} + \boldsymbol{\eta}$

$$\boldsymbol{\eta}(\mathbf{p}, \mathbf{s}, \theta_2, \kappa) \equiv \left(\mathcal{H}(\kappa) \cdot * \tilde{\Delta}(\mathbf{p}, \theta_2) \right)^{-1} \mathbf{q}(\mathbf{p})$$

Challenges:

- ▶ Given $[\mathbf{q}, \mathbf{p}, \tilde{\Delta}, \mathcal{H}(\kappa)]$ I can always produce a vector of marginal costs $\mathbf{mc}$ that rationalizes what we observe. [ie: J equations J unknowns].
- ▶ Nonparametrically we cannot identify κ without more restrictions (!).

What do people do?

Maybe some vectors of $\mathbf{mc}$ look less “reasonable” than others.

- ▶ Marginal costs ≤ 0 seem problematic. [Might just be that your estimates for demand are too inelastic...]
- ▶ or I have a parametric model of MC in mind.

$$f(p_{jt} - \eta_{jt}(\mathbf{p}, \mathbf{s}, \theta_2, \kappa)) = h_s(\mathbf{x}_{jt}, \mathbf{w}_{jt}, \theta_3) + \omega_{jt}$$
$$E[\omega_{jt} | \mathbf{x}_t, \mathbf{w}_t, \mathbf{v}_t] = 0$$

- ▶ Can test that model with GMM objective of mc_{jt} on regressors.
- ▶ Maybe marginal costs cannot deviate too much within product from period to period. (We can write these as moment restrictions too).

Approach #2: Simultaneous Supply and Demand

Estimate θ_2 using both supply and demand. The fit of my supply side will also inform my demand parameters, particularly α the price coefficient. [BLP 95 used this for additional power with lots of random coefficients and potentially weak instruments].

$$\begin{aligned}\delta_{jt}(\mathbf{s}_t, \tilde{\theta}_2) + \alpha p_{jt} &= h_d(\mathbf{x}_{jt}, \mathbf{v}_{jt}, \theta_1) + \xi_{jt} \\ f(p_{jt} - \eta_{jt}(\mathbf{p}, \mathbf{s}, \theta_2, \kappa)) &= h_s(\mathbf{x}_{jt}, \mathbf{w}_{jt}, \theta_3) + \omega_{jt}\end{aligned}$$

Challenges:

- ▶ Should I try to estimate κ ? or just compare objective values at $\kappa_{fg} \in \{0, 1\}$?
- ▶ Am I testing conduct? Or am I testing the functional form for my supply model?
- ▶ Will a missing IV/restriction change whether or not I believe firms are colluding?

There are two ways to think about conduct:

1. Using moment conditions to estimate $\hat{\kappa}$ or $\mathcal{H}(\kappa)$ directly.
 - Often with a small number of parameters (ie: $\kappa_{fg} = 0$ except for firms I know are in a cartel).
 - Can be challenging to tell similar values of κ_{fg} apart (under-powered).
2. “Menu Approach”
 - Nevo (Economics Letters 1998)
 - Bresnahan (1987)
 - Compare some goodness-of-fit criteria across assumed values of κ (Bertrand vs. Collusion)
3. Rejecting (or failing to reject) a single model.

Put the η_{jt} on the RHS and test whether $\lambda = 1$:

$$p_{jt} = h_s(\mathbf{x}_{jt}, \mathbf{w}_{jt}, \theta_3) + \lambda \cdot \eta_{jt}(\mathbf{p}, \mathbf{s}, \theta_2, \kappa) + \omega_{jt} \text{ with } E[\omega_{jt} | \mathbf{x}_t, \mathbf{w}_t, \mathbf{z}_t] = 0$$

- ▶ We are basically running 2SLS with IV for the endogenous η_{jt}
- ▶ “Informal” test of Villas Boas (2007): $\mathbb{E}[\omega_{jt} | \mathbf{x}_{jt}, \mathbf{w}_{jt}, \eta_{jt}] = 0$.
 - Considers different forms of $f(\cdot)$: linear, exponential, logarithmic.
 - Not sure the published paper includes these results (?) WP does?
- ▶ Pakes (2017) uses Wollman (2018) data and BLP IV $\mathbb{E}[\omega_{jt} | x_{jt}, w_{jt}, f(x_{-j})] = 0$.
- ▶ $\lambda \neq 0$ is hard to interpret.

Table 1: Wollman & Pricing Equilibrium.

Taken from Pakes, 2017, *Journal of Industrial Economics*.

	Price (S.E.)		Price (S.E.)	
Gross Weight	.36	(0.01)	.36	(.003)
Cab-over	.13	(0.01)	.13	(0.01)
Compact front	-.19	(0.04)	0.21	(0.03)
long cab	-.01	(0.04)	0.03	(0.03)
Wage	.08	(.003)	0.08	(.003)
<i>Markup</i>	.92	(0.31)	1.12	(0.22)
Time dummies?	No	n.r.	Yes	n.r.
R ²	0.86	n.r.	0.94	n.r.

Note. There are 1,777 observations; 16 firms over the period 1992-2012. S.E.=Standard error.

These are somewhat reassuring:

- ▶ $\lambda \approx 1$ for multiproduct-oligopoly
- ▶ Fit is pretty good $R^2 > 0.8$ and $R^2 > 0.5$ for within vehicle regressions (not shown).
- ▶ As a behavioral model, multiproduct demand estimation seems successful.
- ▶ But, do we know that an alternative $\mathcal{H}(\kappa)$ would have a $\lambda \neq 1$ or a lower R^2 , and if so how low before we can “reject” the model?

Another idea (Bonnet and Dubois, Rand 2010) runs the following regression:

$$\log \left(p_{jt} - \eta_{jt}(\mathbf{p}, \mathbf{s}, \hat{\theta}_2, \kappa) \right) = h_s(x_{jt}, w_{jt}, \theta_3) + \omega_{jt}$$

- ▶ Run a regression for each κ and obtain $Q(\kappa) = \sum_{jt} \hat{\omega}_{jt}^2$
- ▶ Employ the **non nested test** of Rivers and Vuong (2002). Why?
- ▶ Working out the distribution of $Q(\kappa_1) - Q(\kappa_2) = T(\kappa_1, \kappa_2)$ is the hard part.
- ▶ Also this is OLS (or NLLS) and there are no instruments or **exclusion restrictions** for the supply side.
Presumably we could add some and do GMM? (I think this is the “formal” test of Villas Boas (ReStud 2007)).

So far three approaches to exploit $E[\omega_{jt}|x_t, w_t, z_t] = 0$

1. Put the markup on RHS and instrument for it to test $\lambda = 1$ (Wald)

$$p_{jt} = h_s(x_{jt}, w_{jt}, \theta_3) + \lambda \cdot \eta_{jt}(\mathbf{p}, \mathbf{s}, \widehat{\theta}_2, \kappa) + \omega_{jt}$$

2. Put the markup on LHS assuming $\lambda = 1$ and test goodness of fit of supply equation (Anderson Rubin)

$$p_{jt} - \eta_{jt}(\mathbf{p}, \mathbf{s}, \widehat{\theta}_2, \kappa) = h_s(x_{jt}, w_{jt}, \theta_3) + \omega_{jt}$$

3. Estimate supply and demand simultaneously $[\theta_1, \theta_2, \theta_3]$ and compare goodness of fit for different κ . (Likelihood Ratio)

Backus Conlon Sinkinson (2022)

We start with marginal revenue and marginal cost (unobserved ω , observed $h(\cdot)$)

$$\begin{aligned}\psi_{jt}^m &= mc_{jt} \\ p_{jt} - \eta_{jt}^m &= h_s(\mathbf{x}_{jt}, \mathbf{w}_{jt}) + \omega_{jt}^m\end{aligned}$$

- ▶ Let's be vague/flexible with $h_s(\cdot)$ for now, but I don't know the production function.
- ▶ Assume: Demand and hence η_{jt}^m are **known (given conduct)**.
- ▶ Idea (η^A, η^B) are monopoly/perfect competition or Cournot/Bertrand.

The Question

Two competing markups (η_{jt}^A, η_{jt}^B): which fits the data better?
(both may be misspecified)

$$p_{jt} = h_s(\mathbf{x}_{jt}, \mathbf{w}_{jt}) + \tau \eta_{jt}^A + (1 - \tau) \eta_{jt}^B + \omega_{jt}$$

Model is defined by a conditional moment restriction $\mathbb{E}[\omega_{jt} | z_{jt}^s] = 0$

- ▶ $H_0 : \tau = 1$ vs $H_a : \tau = 0$
- ▶ This is a **model selection** problem or a **non nested testing** problem.
 - We might want to compare more than two alternatives (too bad).
- ▶ Obvious endogeneity problem with η_{jt} !

Compare violations of unconditional moments under $(\eta_{jt}^A, \eta_{jt}^B)$ and $A(z_{jt}^s)$:

$$p_{jt} - \eta_{jt}^A = h_s(\mathbf{x}_{jt}, \mathbf{w}_{jt}) + \omega_{jt}^A$$

$$p_{jt} - \eta_{jt}^B = h_s(\mathbf{x}_{jt}, \mathbf{w}_{jt}) + \omega_{jt}^B$$

Which gives us

$$g_A = \frac{1}{N} \sum_{jt} \omega_{jt}^A A(z_{jt}^s), \quad g_B = \frac{1}{N} \sum_{jt} \omega_{jt}^B A(z_{jt}^s)$$

$$Q_m = g'_m W_m g_m$$

Now consider a **Rivers Vuong (2002)** type test $T_{RV} = \sqrt{n} \left(\frac{Q_A - Q_B}{\sigma_{Q_A - Q_B}} \right) \sim N(0, 1)$ $H_0 : Q_A - Q_B = 0$ vs.
 $H_A : Q_A > Q_B$ or $Q_A < Q_B$.

Getting the SD of the difference is hard $\rightarrow$ bootstrap

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- ▶ Bresnahan (1987): Did LR test to determine collusion vs. competition in 1955 automobile price war
 - No IV, errors were measurement in P, Q .
- ▶ Bonnet and Dubois (2010): RV test
 - But no IV – maximum likelihood with normally distributed ω_{jt} 's.
- ▶ Villas Boas (2007): Cox test to determine double marginalization or not in yogurt
 - GMM objective, unclear what if any IV are used.
 - Need to “know” the true model.
- ▶ Duarte, Magnolfi, Solvsten, Sullivan (2022): RV beats Cox pretty badly in Monte Carlo.

Main Results: These are $N(0, 1)$

	Others' Cost	Demographics	BLP Inst.	Dmd. Opt. Inst.
Own Profit Max vs.	Panel 1: $A(\mathbf{z}_t) = \mathbf{z}_t$, linear $h_s(\cdot)$			
Common Ownership	-4.3410	-1.1966	0.5047	-1.2552
Double Marginalization	2.1922	1.0055	-0.0412	7.0897
Double Marginalization + CO	-0.8262	0.6892	0.1428	6.9320
Perfect Competition	3.2995	0.5194	0.7355	3.7223
Monopolist	-2.2264	-1.0528	-0.4525	-0.9202
Own Profit Max vs.	Panel 2: $A(\mathbf{z}_t) = \mathbb{E}[\Delta\eta^{12} \mathbf{z}_t]$, linear $h_s(\cdot)$ and $g(\cdot)$			
Common Ownership	-2.3044	-0.5105	-0.0384	-1.6133
Double Marginalization	0.8644	0.4421	-0.5311	3.3367
Double Marginalization + CO	-0.9382	-0.2389	-0.3684	-0.0045
Perfect Competition	0.7164	0.6135	-0.1080	-0.3151
Monopolist	-0.8577	-0.4002	-0.3868	-1.2339
Own Profit Max vs.	Panel 3: $A(\mathbf{z}_t) = \mathbb{E}[\Delta\eta^{12} \mathbf{z}_t]$, random forest $h_s(\cdot)$ and $g(\cdot)$			
Common Ownership	-3.3777	-3.2509	-3.7130	-4.0256
Double Marginalization	-5.9699	-9.9547	-6.5789	-7.8269
Double Marginalization + CO	-5.9264	-6.1550	-6.5231	-7.4760
Perfect Competition	-4.0468	-6.1901	-5.1494	-6.3484
Monopolist	-3.4972	-4.0070	-3.4358	-3.7495

An Internalization Parameter

Let κ represent the weight a firm places on competitors and τ the internalization of those weights.

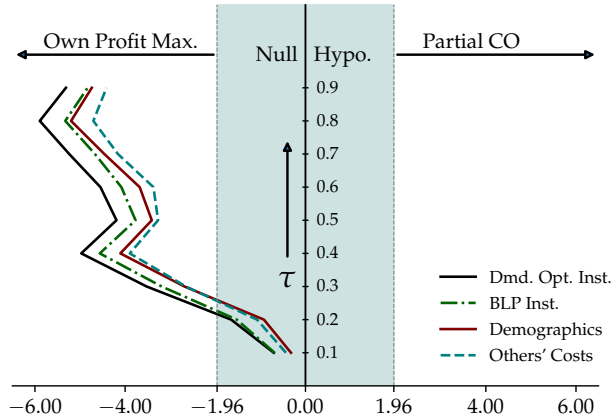
$$\arg \max_{p_j : j \in \mathcal{J}_f} \sum_{j \in \mathcal{J}_f} (p_j - mc_j) \cdot \sigma_j(\mathbf{p}) + \sum_{g \neq f} \tau \kappa_{fg} \sum_{j \in \mathcal{J}_g} (p_k - mc_k) \cdot \sigma_k(\mathbf{p})$$

Now,

- ▶ $\tau = 0$ implies own-profit maximization
- ▶ $\tau = 1$ implies common ownership pricing
- ▶ τ in between is...? Agency?

We test $\tau \in (0.1, \dots, 0.9)$ against own-profit maximization.

Internalization Parameter Results



The true model for markups (conduct) will satisfy the CMR: $\mathbb{E}[\omega_{jt}|z_{jt}^s] = 0$

$$p_{jt} - \eta_{jt}^{(m)} = h_s(x_{jt}, w_{jt}; \theta_3) + \omega_{jt}$$

Goal is test two competing markups $\eta_{jt}^{(A)}, \eta_{jt}^{(B)}$, but there are challenges:

1. Test will depend on how we choose **unconditional moment restrictions** $\mathbb{E}[\omega_{jt} \cdot A(z_{jt}^s)] = 0$
2. Test may depend on how we specify $h_s(\cdot)$
 - All tests are basically joint tests of the specification for **observed marginal costs** and the **exclusion restriction**.
 - Villas Boas (2007) tries log, linear, exponential in $x\beta$
3. Choice of $\eta_{jt}^{(m)}$ will affect our choice of **weighting matrix** and thus the test. (Hall Pelletier (2011))

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A Brief Aside: Chamberlain (1987) in a Slide

What contains as much information as the CMR $\mathbb{E}[\omega|z_{jt}^s]$ and moments of the form $\mathbb{E}[\omega_{jt} \cdot A(z_{jt}^s)]$.

- ▶ For linear models $A(z_{jt}^s) = z_{jt}^s$ is generally without loss.
- ▶ For nonlinear models, Chamberlain (1987) shows that the efficient estimator uses

$$A(z_{jt}^s) = \mathbb{E} \left[\frac{\partial \omega_{jt}}{\partial \theta} | z_{jt}^s \right]$$

- ▶ That is not too helpful (its a function of the unknown θ).
- ▶ Much of the follow-up work has been about feasible approximations to this “optimal instrument” (e.g., Newey 1990)

For us a similar concern arises, but it is about **power** to distinguish conduct models rather than **efficiency** of estimation.

Our Idea: Motivation #1 (Optimal IV)

The model is given by

$$p_{jt} = h_s(\mathbf{x}_{jt}, \mathbf{w}_{jt}, \theta_3) + \tau \cdot \eta_{jt}^A + (1 - \tau) \cdot \eta_{jt}^B + \omega_{jt}^m$$

where $H_0 : \tau = 1$ and $H_a : \tau = 0$

- ▶ The optimal IV in the Chamberlain (1987) sense is given by $\mathbb{E} \left[\frac{\partial \omega_{jt}}{\partial \tau} | z_t \right] = \mathbb{E} [\eta_{jt}^A - \eta_{jt}^B | z_t]$.
- ▶ In words: The IV need to predict the **difference in markups** (beyond observed $h_s(\mathbf{x}_{jt}, \mathbf{w}_{jt}, \theta_3)$).

Our Idea: Motivation #2 (Misspecification)

Index the **true** model by 0. Then,

$$p_{jt} - \eta_{jt}^0 = h_s(x_{jt}, w_{jt}) + \omega_{jt}^0.$$

To motivate a useful test, we ask what happens when we estimate supply with the **wrong** conduct model (1):

$$p_{jt} - \eta_{jt}^1 = h_s(x_{jt}, w_{jt}) + \underbrace{\underbrace{\eta_{jt}^0 - \eta_{jt}^1}_{\equiv \Delta \eta_{jt}^{0,1}} + \omega_{jt}^0}_{\omega_{jt}^1}.$$

- ▶ Misspecifying conduct introduces an omitted variable: the difference in markups.
- ▶ Our test is premised on detection of this omitted variable.

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$$p_{jt} - \eta_{jt}^1 = h_s(x_{jt}, w_{jt}) + \underbrace{\eta_{jt}^0 - \eta_{jt}^1 + \omega_{jt}^0}_{\omega_{jt}^1}.$$

- ▶ Misspecifying conduct introduces an omitted variable: the difference in markups.
- ▶ Our test is premised on detection of this omitted variable.

Our Innovation: How does this help?

The model is given by

$$p_{jt} - \eta_{jt}^m = h_s(\cdot) + \omega_{jt}^m, \text{ and } \mathbb{E}[\omega_{jt}^{(m)} \cdot A(z_t)] = 0.$$

We suggest $A(z_t) = \mathbb{E}[\eta_{jt}^1 - \eta_{jt}^2 | z_t]$; several advantages:

- ▶ Reduces potentially many moments ($\mathbb{E}[\omega_{jt}' z_t] = 0$) to a single, scalar moment. No need for a weighting matrix, or associated problems.
- ▶ Testing is reduced to two prediction exercises: $\mathbb{E}[\eta_{jt}^1 - \eta_{jt}^2 | z_t]$ and $\hat{\omega}_{jt}^{(m)}$.
- ▶ Show in the paper that this leads to the most powerful test (maximizes distance between two GMM objective functions conditional on weight matrix).
- ▶ Downside: Our choice of instrument is **model specific**! UMP is not going to happen.

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Possible Exclusion Restrictions

We are looking for variables which affect **demand but not supply**:

$$\sigma_j^{-1}(\mathbf{s}_t, \mathbf{p}_t, \mathbf{y}_t, \mathbf{x}_t, \mathbf{v}_t, \tilde{\theta}_2) = h_d(\mathbf{x}_{jt}, \mathbf{v}_{jt}; \theta_1) - \alpha p_{jt} + \lambda \log(\text{ad}_{jt}) + \xi_{jt}$$
$$p_{jt} - \eta_{jt}(\mathbf{s}_t, \mathbf{p}_t; \theta_2, \mathcal{H}_t(\kappa)) = h_s(\mathbf{x}_{jt}, \mathbf{w}_{jt}; \theta_3) + \omega_{jt}$$

Things we use:

- Obvious choice: $\mathbf{v}_{jt}$ (things like product recalls are relatively weak)
- Demographics (enter nonlinearly): $\mathbf{y}_t$ (chain-level income works well)
- Characteristics of other goods: $f(\mathbf{x}_{-j,t})$ (BLP instruments).
- Characteristics of other goods: $\mathbf{w}_{-j,t}$ (commodity price of oats for Rice Krispies)

Things we don't use:

- Unobserved demand shocks ξ_{jt} (see MacKay Miller 2020 for $Cov(\xi_j, \omega_j) = 0$).
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Algorithm

(1) Split the sample by markets t into 70% *test* and 30% *train*.

(2) On the *training sample*:

(a) Approximate the optimal instruments $a(z_{jt}^s) = \mathbb{E}[\Delta\eta_{jt}^{(1,2)} \mid z_{jt}^s]$ as the fitted values from:

$$\Delta\eta_{jt}^{1,2} = g(z_{jt}^s) + \zeta_{jt}.$$

(b) Estimate the marginal cost function, under models 1 and 2 to obtain residuals $\widehat{\omega}_{jt}^1$ and $\widehat{\omega}_{jt}^2$:

$$p_{jt} - \eta_{jt}^m = h_s(\mathbf{x}_{jt}, \mathbf{w}_{jt}; \theta_3) + \omega_{jt}^m.$$

(3) On the *test sample*:

(a) For each candidate model, compute the value of the scalar moment:¹

$$Q(\eta^m) = \left(\sum_{j,t} \widehat{\omega}_{jt}^m \cdot \widehat{g}(\mathbf{z}_t) \right)^2.$$

(b) Repeat the previous step on bootstrapped samples and estimate $\widehat{\sigma}/\sqrt{n}$ the standard error of the difference $\tilde{Q}(\eta^1) - \tilde{Q}(\eta^2)$.

(c) Compute the test statistic

$$T = \frac{\sqrt{n}(Q(\eta^1) - Q(\eta^2))}{\widehat{\sigma}} \sim \mathcal{N}(0, 1).$$

Note: Steps 2(a) and 2(b) can be done in any order via non-parametric regression.

Limitations

Not everything is testable:

- ▶ If $\Delta\eta_{jt}$ cannot be explained by z_{jt}^s beyond contents of (x_j, w_j) we have nothing
- ▶ Flexible demand models are required to generate cross sectional variation in markups
- ▶ Beware of “accidental” exclusion restrictions.

Alternate Justifications

1. Model Misspecification: if one of the two models is correct, $\eta_{jt}^1 - \eta_{jt}^2$ exactly corresponds to the misspecification error when using the other. [▶ detail](#).
2. Difference in Test Statistics: We can also show it maximizes the difference in GMM objectives $(Q_1 - Q_2)$. (Which is almost power for fixed σ). Also we need that either $h_s(\cdot)$ doesn't depend on $\eta^{(m)}$ or that it is linear so we can residualize on (x, w) .